Anthropometry and Growth Charts

- ☑ Anthropometry should include weight, standing height (stadiometer) length by infantometer (<2 years), proportion (upper segment : lower segment), head circumference, and arm span. Measuring forearm length gives an important clue for SHOX mutation (short forearm)
- ☑ The combined World Health Organization/Indian Academy of Pediatrics (WHO/IAP) growth charts can be used in office practice. 2006 WHO Multicenter Growth Reference Study (MGRS) growth charts for children from birth to 5 years. Revised IAP 2015 growth charts for children from 5 to 18 years. Height age and weight age can be plotted on the chart, weight age is more affected in chronic diseases whereas height age is more affected in endocrinopathies. Short stature with obesity is almost always pathological.
- ☑ Plotting MPH will help in differentiating familial and constitutional short stature. Bone age and puberty are delayed in CDGP and is more common in boys. In familial short stature (FSS), bone age and pubertal timing is normal and there is a family history of short stature. Final adult height is normal in CDGP.

Investigations

First-line investigations	***Second-line investigations***
Complete hemogram/erythrocyte sedimentation rate (ESR)/C-reactive protein (CRP)	GH stimulation tests
Renal and liver function tests	MRI brain (to rule out pituitary/ congenital malformations)
Venous blood gas/electrolytes	Genetic tests for GHD/MPHD/ syndromes
Calcium, phosphorous, alkaline phosphatase (ALP), and parathyroid hormone (PTH)	
Thyroid function tests	
Celiac screen anti-tissue transglutaminase immunoglobulin A (Tg-IgA)	
Insulin-like growth factor-1 (GF-1) [insulin-like growth factor-binding protein 3 (IGFBP3) in <2 years children and malnutrition]	
Karyotyping in all girls with unexplained short stature	

Growth hormone stimulation tests—at least two growth provocation tests should be done in suspected isolated GHD [clonidine/arginine/insulin/glucagon/dopamine/ growth hormone-releasing hormone (GHRH)] before considering GH treatment. One stimulation test is enough in diagnosed intracranial lesions, MPHD, history of irradiation, and genetic defects.

Evaluation

Radiological Investigations

- ☑ Hand and wrist X-ray posteroanterior (PA) view of the nondominant hand is taken and bone age is assessed by any one of the standard methods. Bone age is usually delayed by >2 years than CA in pathological causes and within 2 years in CDGP. Hand X-ray also gives insight into rickets, Turner, and SHOX mutation. A skeletal survey (X-ray skull, spine, and long bones) is required in cases of disproportionate short stature.
- ☑ MRI brain—neuroimaging is done to assess anterior pituitary dimensions, pituitary stalk, posterior pituitary, exclude tumors, and malformations associated with hypopituitarism such as optic nerve hypoplasia, holoprosencephaly, and corpus callosum agenesis.
- ☑ Genetic testing—reserved for specific syndromes.

Management

- ☑ If there is a definite cause for the short stature, treatment is directed toward the cause. When no cause is apparent, child should be followed 6 monthly to monitor growth velocity and a faltering growth curve warrants more investigations.
- ☑ Reassurance of the family in case of FSS and constitutional delay.

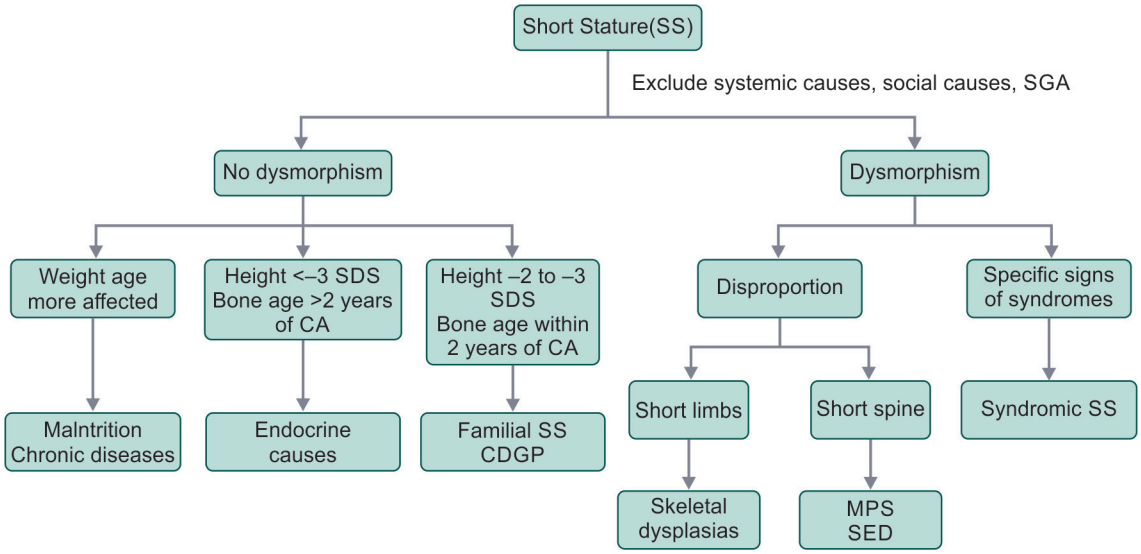
Specific Management

Primary hypothyroidism	Thyroxine
MPHD	Glucocorticoids, thyroxine, GH, and sex steroids
Malnutrition	Nutritional rehabilitation
Rickets	Vitamin D
Psychosocial	Good social environment and care
Chronic systemic diseases	Address the cause
CDGP	Short course of androgens is helpful (depot testosterone)

Growth Hormone: Indications and Criteria

GHD	☑ Severe short stature (height >3 SD below mean) ☑ Height >1.5 SD below MPH ☑ Height >2 SD below mean and height velocity >1 SD below mean ☑ Signs of MPHD/intracranial lesion
Chronic renal disease	Response is better if started in early chronic kidney disease (CKD) than end-stage renal disease (ESRD)
Turner syndrome	GH initiation at least 4 years prior to pubertal induction gives good outcome
SGA infants	☑ Birth weight or length >2 SD below mean ☑ Height < –2.5 SD of mean at start of therapy ☑ Current height >1 SD below MPH ☑ Failure to catch up by 4 years
Prader–Willi syndrome	Genetic diagnosis is mandatory before starting GH
Idiopathic short stature	Height >2.25 SD below mean

Flowchart for Short Stature in Adolescent



(MPS: mucopolysacchridosis; SED: spondylo ephphyseal dysplasia)

Psychosocial Effects of Short Stature in Adolescents

- ☑ Juvenilization—social isolation and discrimination
- ☑ Bullying in school
- ☑ Low self-esteem
- ☑ Behavioral problems both internalizing and externalizing

Psychosocial Interventions

Problem	*Strategy*	*Efficacy*
Juvenilization	Reinforce independence/self-help	No systematic studies
Bullying	☑ Cognitive restructuring ☑ Coping strategies	Possibly efficacious
Generalized anxiety	Cognitive behavioral therapy (CBT)	Probably efficacious
Depression	☑ CBT ☑ Interpersonal psychotherapy	Well established

Further Reading

- ☑ Indian Academy of Pediatrics. Growth Assessment and Growth Disorders. Mumbai, India: Indian Academy of Pediatrics; 2016.
- ☑ Khadilkar V, Bajpai A, Prasad KH. IAP Textbook on Pediatric Endocrinology, 1st edition. New Delhi: Jaypee Brothers Medical Publishers; 2019.
- ☑ Mergulhão B, Almeida JP, Moreira H, Castro-Correia C, Bullinger M, Canavarro MC, et al. Quality of life of children and adolescents with short stature: the twofold contribution of physical growth and adaptive height-related cognitive beliefs. J Clin Psychol Med Settings. 2022;29:466-75.
- ☑ Sandberg DE, Voss LD. The psychosocial consequences of short stature: a review of the evidence. Best Pract Res Clin Endocrinol Metab. 2002;16(3):449-63.
- ☑ Yadav S, Dabas A. Approach to short stature. Indian J Pediatr. 2015;82(5):462-70.

Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Feeding in Preterm and Feed Intolerance

Lead Author

Naveen Bajaj

Co-Authors

Bijan Saha, Giridhar Sethuraman

Under the Auspices of the IAP Action Plan 2022

Remesh Kumar R
IAP President 2022

Upendra Kinjawadekar
IAP President-Elect 2022

Piyush Gupta
IAP President 2021

Vineet Saxena
IAP HSG 2022–2023

© Indian Academy of Pediatrics

IAP Standard Treatment Guidelines Committee

Chairperson

Remesh Kumar R

IAP Coordinator

Vineet Saxena

National Coordinators

SS Kamath, Vinod H Ratageri

Member Secretaries

Krishna Mohan R, Vishnu Mohan PT

Members

Santanu Deb, Surender Singh Bisht, Prashant Kariya, Narmada Ashok, Pawan Kalyan